NEET UG PHYSICS STANDARD

Mechanical Waves: Propagation, Superposition, Normal Modes, and Beats

A comprehensive, self-contained study module covering progressive waves, wave speed in strings and fluids, Newton's formula and Laplace correction, interference, reflection, standing waves in strings and organ pipes, and beat phenomena.

1. Learning Objectives & Syllabus Overview

After studying this tutorial, you will be able to:

- Distinguish between mechanical, electromagnetic, and matter waves, and explain why waves transport energy and information without net bulk flow of matter.
- Classify mechanical waves into transverse and longitudinal waves based on particle oscillation direction and the elastic stress sustained by the medium.
- Analyze the mathematical displacement relation $y(x,t) = a\sin(kx - \omega t + \phi)$ for harmonic progressive waves and extract amplitude, wavelength, angular wave number, time period, frequency, and wave velocity.
- Calculate the speed of transverse waves on a stretched string $v = \sqrt{T/\mu}$ and longitudinal waves in fluids and solids, applying Laplace's adiabatic correction to Newton's formula for gases.
- Apply the principle of superposition to explain constructive and destructive interference, phase reversal at rigid boundaries, and reflection without phase change at open boundaries.
- Formulate the equations for stationary (standing) waves, locate nodes and antinodes, and compute the normal mode frequencies (harmonics and overtones) for stretched strings, closed organ pipes, and open organ pipes.
- Solve numerical and conceptual NEET UG problems involving beat frequency $\nu_{\text{beat}} = |\nu_1 - \nu_2|$ and string tuning adjustments.

2. Core Concepts and Detailed Explanations

2.1 Introduction to Wave Motion

When an object oscillates in isolation, its energy remains confined to that single oscillator. However, in a material medium, constituent particles (atoms, molecules, or macroscopic elements) are bound to one another by elastic intermolecular forces. Consequently, the oscillation of one particle disturbs its immediate neighbours, which in turn disturb their neighbours.

Consider dropping a small pebble into a pond of still water. Circular ripples spread outward from the point of impact. If cork pieces are floating on the water surface, they merely bob up and down about their mean positions without flowing outward with the expanding circles. Similarly, when we speak, sound travels outward through the air to a listener's ear or a microphone without any net wind or bulk flow of air from the speaker to the listener.

Definition of a Wave: A wave is a propagating pattern of disturbance that transfers energy, momentum, and information from one point to another without the actual physical transfer or bulk flow of matter as a whole.

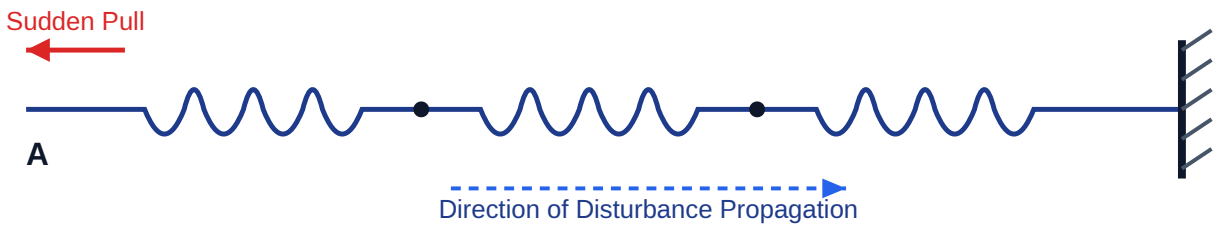
Classification of Waves

1. **Mechanical Waves:** Waves that require a material medium possessing both *elasticity* (to provide a restoring force) and *inertia* (mass density, enabling particles to overshoot equilibrium and oscillate). Examples include waves on a stretched string, water waves, sound waves, and seismic waves. They cannot travel through vacuum and are governed by Newton's laws of motion.
2. **Electromagnetic Waves:** Waves consisting of coupled oscillating electric and magnetic fields that do not require a material medium for propagation. Examples include visible light, radio waves, microwaves, and X-rays. In vacuum, all electromagnetic waves travel at the exact same speed c:
$$c = 299,792,458\,\text{m}\,\text{s}^{-1} \approx 3 \times 10^{8}\,\text{m}\,\text{s}^{-1}$$
3. **Matter Waves:** Waves associated with moving microscopic constituents of matter such as electrons, protons, neutrons, atoms, and molecules. Described by quantum mechanics, matter waves associated with high-speed electrons are practically employed in electron microscopes.

BOSONANCE

Spring-Coupling Model of Mechanical Wave Propagation

To visualize how elastic forces transmit a disturbance through a medium, consider a line of identical springs connected end-to-end with one extreme end attached to a rigid support. If the free end A is suddenly pulled and released, the first spring stretches from its equilibrium length. Because it is coupled to the second spring, it pulls the second spring, which then stretches the third, and so on. The disturbance travels from one end of the chain to the other, while each individual spring executes only small harmonic oscillations about its own equilibrium position. A practical macroscopic analogy is a stationary train at a railway station: when an engine gently pushes the first bogie, the spring coupling transmits the push sequentially from bogie to bogie without the entire train being bodily displaced at once.



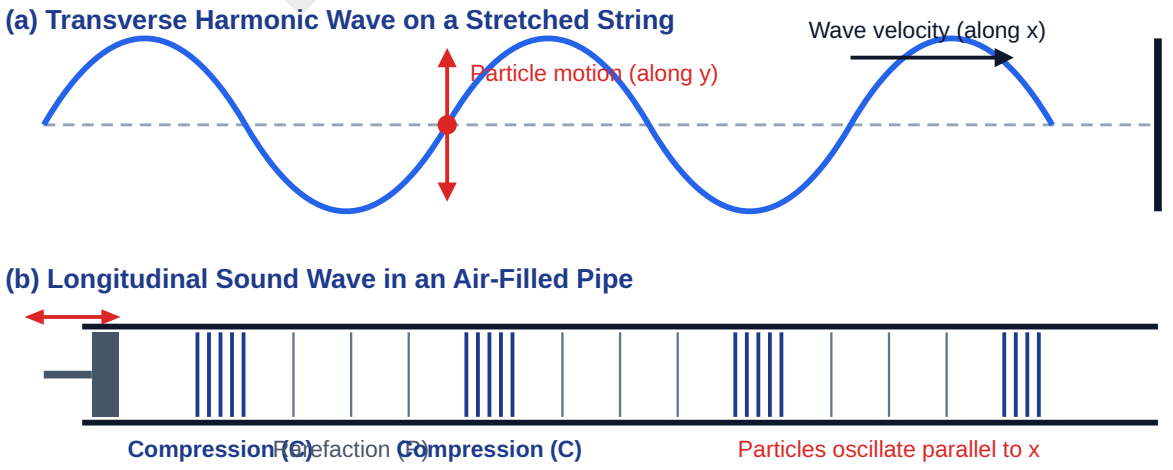
Tutorial Diagram 1: A chain of coupled springs. Pulling end A disturbs the first spring, and the elastic coupling transmits the disturbance rightward to the fixed wall.

In air, sound waves propagate by a similar mechanism. When a small region of air is compressed, its mass density increases by $\delta\rho$, inducing an increase in local pressure by δp. Because pressure is force per unit area, this excess pressure provides a restoring force proportional to the density disturbance. Compressed air molecules push out into the adjoining region, compressing that next layer while leaving the original region rarefied (expanded, lower density). Surrounding air then rushes into the rarefied zone, causing the rarefaction to shift outward as well. In crystalline solids, atoms arranged in a periodic lattice experience interatomic restoring forces when displaced from equilibrium, acting like masses connected by three-dimensional springs.

2.2 Transverse and Longitudinal Waves

Mechanical waves are categorized into two fundamental types depending on the direction of oscillation of the medium's constituents relative to the direction of wave propagation:

- **Transverse Waves:** The constituents of the medium oscillate *perpendicular (normal)* to the direction of wave propagation. If a horizontal stretched string is given a single up-and-down jerk at one end, a single transverse *pulse* travels along its length. If the end is driven continuously with a periodic sinusoidal up-and-down motion, a sinusoidal (harmonic) transverse wave travels along the string.
- **Longitudinal Waves:** The constituents of the medium oscillate *along (parallel or anti-parallel to)* the direction of wave propagation. If a piston at one end of a long air-filled pipe is oscillated forward and backward sinusoidally, alternating regions of **condensations** (compressions, high density and pressure) and **rarefactions** (expansions, low density and pressure) travel along the pipe.



Tutorial Diagram 2: Comparison of (a) transverse wave where string elements oscillate perpendicular to propagation, and (b) longitudinal sound wave where air layers oscillate parallel to propagation.

BOSONANCE

Elastic Media Requirements & Special Cases

- **Why Transverse Waves Require Solids (or Tensioned Surfaces):** In a transverse wave, adjacent layers of the medium slide past one another normal to the direction of wave travel, subjecting each element to *shearing strain*. Therefore, transverse mechanical waves can propagate only in media that can sustain **shearing stress** (i.e., media possessing a shear modulus of rigidity, such as solids, or stretched strings under tension). Fluids (liquids and gases) have no shape of their own and cannot sustain shearing stress inside their bulk; hence, bulk fluids cannot propagate transverse waves.
- **Why Longitudinal Waves Propagate in All Elastic Media:** Longitudinal waves involve local volume changes (compressive and tensile volumetric strain). Because solids, liquids, and gases all possess a **bulk modulus of elasticity** (or Young's modulus in solid rods) and can sustain compressive stress, longitudinal waves propagate through all three states of matter. In a medium like steel, both transverse and longitudinal waves can propagate (and they generally travel at different speeds in the same medium).
- **Surface Water Waves (Capillary and Gravity Waves):** Waves on the surface of water are of two types:
  1. *Capillary waves:* Ripples of short wavelength (not more than a few centimetres) whose restoring force is provided by the **surface tension** of water.
  2. *Gravity waves:* Longer waves with wavelengths ranging from several metres to several hundred metres, where the restoring force is the **pull of gravity** tending to keep the water surface at its lowest level. Particle oscillations in gravity waves are not confined to the surface; they extend with diminishing amplitude all the way to the bottom.

  In water waves (such as ocean waves), water particles move both up-and-down and back-and-forth (tracing elliptical/circular paths). Thus, water waves are a **combination of both transverse and longitudinal waves**.

2.3 Displacement Relation in a Progressive Wave

A wave that travels continuously from one region of a medium to another is called a **travelling wave** or **progressive wave**. To describe it mathematically, we require a function of both spatial position x and time t. For a sinusoidal (harmonic) wave travelling along the positive x-direction, the transverse displacement $y(x,t)$ is given by:

$$y(x,t) = a\sin(kx - \omega t + \phi)$$

For a longitudinal progressive wave, the same mathematical structure applies, except the transverse displacement $y(x,t)$ is replaced by the longitudinal displacement $s(x,t)$ parallel to the wave propagation direction:

$$s(x,t) = a\sin(kx - \omega t + \phi)$$

Including the initial phase angle ϕ is mathematically equivalent to expressing the wave as a linear combination of sine and cosine terms:

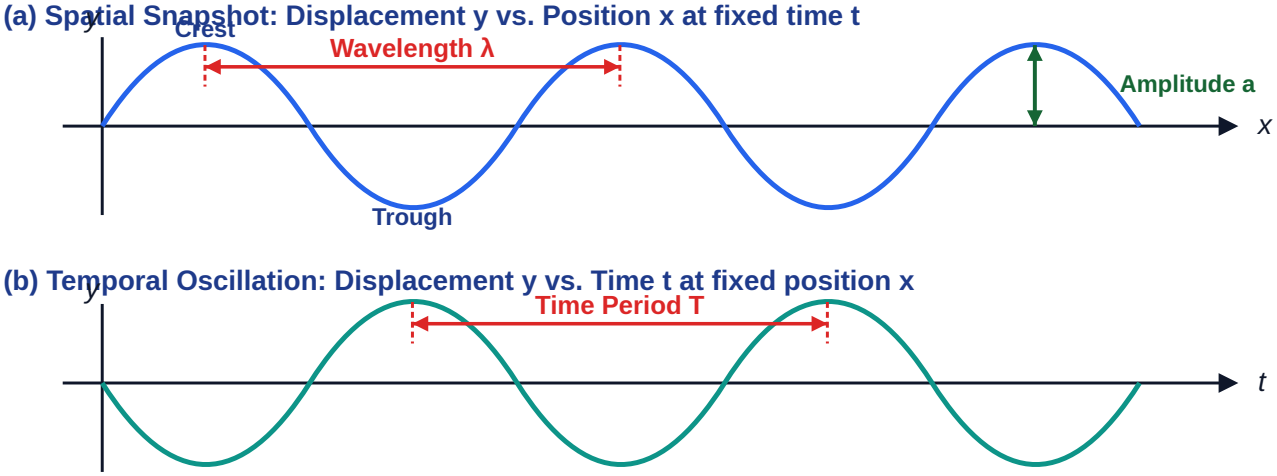
$$y(x,t) = A\sin(kx - \omega t) + B\cos(kx - \omega t)$$

where the resultant amplitude is $a = \sqrt{A^2 + B^2}$ and the initial phase angle is $\phi = \tan^{-1}\left(\frac{B}{A}\right)$.

Direction of Propagation: Positive vs. Negative x-Axis

- **Wave travelling in the $+x$ direction:** $y(x,t) = a\sin(kx - \omega t + \phi)$ or $a\sin(\omega t - kx + \phi')$. Here, x and t have *opposite signs*. To keep the phase $(kx - \omega t + \phi)$ constant as time t increases, position x must increase.
- **Wave travelling in the $-x$ direction:** $y(x,t) = a\sin(kx + \omega t + \phi)$. Here, x and t have the *same sign*. As time t increases, x must decrease to keep the phase constant.

BOSONANCE



Tutorial Diagram 3: Two complementary views of a progressive wave: (a) fixing time t reveals the spatial wavelength λ; (b) fixing location x reveals the temporal period T of simple harmonic motion.

Physical Parameters of a Harmonic Wave

1. **Amplitude (a):** The magnitude of maximum displacement of a particle of the medium from its equilibrium position. While instantaneous displacement $y(x,t)$ oscillates between $+a$ and $-a$, the amplitude a is always a positive scalar quantity. A point of maximum positive displacement ($+a$) is a **crest**; a point of maximum negative displacement ($-a$) is a **trough**.
2. **Phase and Initial Phase (ϕ):** The total argument $(kx - \omega t + \phi)$ of the oscillatory function is the *phase* of the wave, which determines the state of displacement and motion at any (x,t). At $x = 0$ and $t = 0$, the phase equals ϕ, called the **initial phase angle** or phase constant.
3. **Wavelength (λ) and Angular Wave Number (k):** Wavelength λ is the minimum distance between two points having the same phase at a given instant (for instance, between two consecutive crests or two consecutive troughs). At $t = 0$ (with $\phi = 0$), $y(x,0) = a\sin(kx)$. Since $\sin(kx) = \sin(kx + 2n\pi) = \sin\left[k\left(x + \frac{2n\pi}{k}\right)\right]$ for $n = 1, 2, 3, \ldots$, the smallest spatial repetition distance ($n = 1$) is $\lambda = \frac{2\pi}{k}$. Thus:
$$k = \frac{2\pi}{\lambda}$$
Here k is called the **angular wave number** or **propagation constant**. Its SI unit is $\mathrm{rad\,m^{-1}}$ (or simply $\mathrm{m^{-1}}$). It represents the phase difference per unit distance along the wave.
4. **Time Period (T), Angular Frequency (ω), and Frequency (ν):** Fixing position at $x = 0$ (with $\phi = 0$) gives $y(0,t) = a\sin(-\omega t) = -a\sin(\omega t)$. The time required for an element to complete one full oscillation is the time period T. Since $-a\sin(\omega t) = -a\sin[\omega(t+T)]$ requires $\omega T = 2\pi$:
$$\omega = \frac{2\pi}{T} = 2\pi\nu \quad \text{and} \quad \nu = \frac{1}{T} = \frac{\omega}{2\pi}$$
The SI unit of angular frequency ω is $\mathrm{rad\,s^{-1}}$, and the SI unit of frequency ν is hertz (Hz or $\mathrm{s^{-1}}$).

2.4 The Speed of a Travelling Wave

As a wave progresses, any point of fixed phase (such as a crest) shifts by a distance Δx in a small time interval Δt. Since the phase along the trajectory of this crest remains constant:
$$kx - \omega t = \text{constant} \implies k\Delta x - \omega\Delta t = 0$$
In the limit of vanishingly small Δx and Δt, the phase speed (wave speed) v is:

$$v = \frac{dx}{dt} = \frac{\omega}{k} = \frac{2\pi\nu}{2\pi/\lambda} = \nu\lambda = \frac{\lambda}{T}$$

This fundamental relation holds for all progressive waves: in the time T taken by a medium particle to complete one oscillation, the wave pattern advances by a distance equal to one wavelength λ.

Crucial NEET Concept — Wave Speed vs. Particle Speed:

- **Wave Speed ($v = \omega/k = \nu\lambda$):** Constant for a homogeneous medium under fixed physical conditions (temperature, tension). It depends *only* on the elastic and inertial properties of the medium, *not* on the frequency, wavelength, or velocity of the source. The source fixes the frequency ν; the medium fixes the wave speed v; wavelength is then determined via $\lambda = v/\nu$.
- **Particle Velocity ($v_p = \frac{\partial y}{\partial t}$):** The oscillatory velocity of a constituent particle about its equilibrium position:

$$v_p(x,t) = \frac{\partial y}{\partial t} = -a\omega\cos(kx - \omega t + \phi)$$

The maximum particle speed is $(v_p)_{\max} = a\omega$. Notice that $\frac{\partial y}{\partial t} = -\frac{\omega}{k}\frac{\partial y}{\partial x} = -v\frac{\partial y}{\partial x}$, meaning $v_p = -v \times (\text{slope of wave profile})$.

2.4.1 Speed of a Transverse Wave on a Stretched String

For a stretched string, the restoring force is provided by the tension T (dimensions $[\mathrm{MLT}^{-2}]$), and the inertial property is the **linear mass density** $\mu = \frac{m}{L}$ (mass per unit length, dimensions $[\mathrm{ML}^{-1}]$). Combining T and μ dimensionally gives $\frac{[T]}{[\mu]} = \frac{[\mathrm{MLT}^{-2}]}{[\mathrm{ML}^{-1}]} = [\mathrm{L}^2\mathrm{T}^{-2}]$, which is the dimension of v^2. With the exact dimensionless constant $C = 1$ from Newtonian mechanics, the speed of transverse waves on a stretched string is:

$$v = \sqrt{\frac{T}{\mu}}$$

If a wire of radius r and material density ρ has cross-sectional area $A = \pi r^2$, then $\mu = \frac{\rho A L}{L} = \rho A = \rho\pi r^2$, so $v = \sqrt{\frac{T}{\rho A}}$.

2.4.2 Speed of a Longitudinal Wave (Speed of Sound)

In a fluid or bulk medium, longitudinal sound waves propagate via compressions and rarefactions. The relevant elastic modulus is the **bulk modulus** B, and the inertial property is the **volumetric mass density** ρ (dimensions $[\mathrm{ML}^{-3}]$):

$$B = -\frac{\Delta P}{\Delta V/V}$$

where ΔP is the pressure change producing volumetric strain $\Delta V/V$. Combining B ($[\mathrm{ML}^{-1}\mathrm{T}^{-2}]$) and ρ ($[\mathrm{ML}^{-3}]$) yields:

$$v = \sqrt{\frac{B}{\rho}} \quad (\text{in fluids / bulk media})$$

For a one-dimensional solid rod or metallic bar where lateral expansion is negligible, the bar undergoes longitudinal strain governed by **Young's modulus** Y:

$$v = \sqrt{\frac{Y}{\rho}} \quad (\text{longitudinal wave in a solid bar})$$

Why sound travels faster in solids and liquids than in gases: Although solids and liquids have much higher mass densities (ρ) than gases, they are far harder to compress, meaning their elastic moduli (B or Y) are larger by a much greater factor. Thus, the ratio B/ρ (or Y/ρ) is much larger in solids and liquids than in gases.

BOSONANCE

Reference Speeds of Sound in Various Media

Category	Medium	Speed of Sound ($\mathrm{m\,s^{-1}}$)
Gases	Air ($0\,^\circ\mathrm{C}$)	331
	Air ($20\,^\circ\mathrm{C}$)	343
	Helium	965
	Hydrogen	1284
Liquids	Water ($0\,^\circ\mathrm{C}$)	1402
	Water ($20\,^\circ\mathrm{C}$)	1482
	Seawater	1522
Solids (Longitudinal)	Aluminium	6420
	Granite	6000
	Steel	5941
	Copper	3560
	Vulcanised Rubber	54

Newton's Formula and Laplace Correction for Ideal Gases

1. **Newton's Isothermal Assumption:** An ideal gas obeys $PV = Nk_BT$. Newton assumed that the pressure and volume variations during the passage of a sound wave occur slowly enough for the temperature T to remain constant (*isothermal process*). Differentiating $PV = \text{constant}$ gives:

$$V\Delta P + P\Delta V = 0 \implies B_{\text{iso}} = -\frac{\Delta P}{\Delta V/V} = P$$

   Substituting $B = P$ gives **Newton's Formula**:

$$v = \sqrt{\frac{P}{\rho}}$$

   At STP ($P = 1.01 \times 10^5\,\mathrm{N\,m^{-2}}$, $\rho = 1.29\,\mathrm{kg\,m^{-3}}$), Newton's formula predicts $v \approx 280\,\mathrm{m\,s^{-1}}$, which is about 15% lower than the experimental value of $331\,\mathrm{m\,s^{-1}}$.

2. **Laplace's Adiabatic Correction:** Laplace pointed out that compressions and rarefactions in sound waves occur so rapidly—and gases have such low thermal conductivity—that there is virtually no time for heat to flow between adjacent layers. Hence, the process is **adiabatic**, satisfying $PV^\gamma = \text{constant}$, where $\gamma = C_p/C_v$ is the ratio of specific heats. Differentiating $PV^\gamma = \text{constant}$ yields:

$$P\gamma V^{\gamma-1}\Delta V + V^\gamma \Delta P = 0 \implies B_{\text{ad}} = -\frac{\Delta P}{\Delta V/V} = \gamma P$$

   Substituting $B_{\text{ad}} = \gamma P$ gives the **Laplace-corrected formula**:

$$v = \sqrt{\frac{\gamma P}{\rho}} = \sqrt{\frac{\gamma RT}{M}}$$

   For diatomic air, $\gamma = 7/5 = 1.4$. At STP, $v = \sqrt{1.4} \times 280 \approx 331.3\,\mathrm{m\,s^{-1}}$, in excellent agreement with experiment!

Effect of Environmental Factors on Speed of Sound in Air

- **Effect of Pressure (at constant temperature):** Since $P/\rho = RT/M$ is constant at a fixed temperature T, changing the pressure P changes the density ρ in exact proportion. Therefore, *the speed of sound in an ideal gas is independent of pressure at constant temperature*.
- **Effect of Temperature:** Since $v = \sqrt{\frac{\gamma RT}{M}}$, the speed of sound is directly proportional to the square root of the absolute temperature: $v \propto \sqrt{T}$.
- **Effect of Humidity:** Moist air contains water vapour ($\mathrm{H_2O}$, molar mass $M \approx 18\,\mathrm{g\,mol^{-1}}$) replacing dry air molecules ($\mathrm{N_2}$ and $\mathrm{O_2}$, average molar mass $M \approx 29\,\mathrm{g\,mol^{-1}}$). Consequently, the density ρ of moist air is *less* than that of dry air at the same pressure and temperature. Since $v = \sqrt{\gamma P/\rho}$, *the speed of sound increases with humidity*.

BOSONANCE

2.5 The Principle of Superposition of Waves and Interference

When two or more wave pulses travelling through a medium overlap in space and time, each wave pulse continues to retain its individual identity after crossing the others. During the interval of overlap, the net displacement of any medium particle at any position and instant is the **algebraic sum** of the displacements produced by each wave individually:

$$y(x,t) = y_1(x,t) + y_2(x,t) + \cdots + y_n(x,t) = \sum_{i=1}^{n} f_i(x - vt)$$

Interference of Two Co-propagating Harmonic Waves

Consider two sinusoidal waves of equal amplitude a, angular frequency ω, and wave number k, both travelling along the $+x$ direction on a stretched string, differing only by an initial phase constant ϕ:

$$y_1(x,t) = a\sin(kx - \omega t) \quad \text{and} \quad y_2(x,t) = a\sin(kx - \omega t + \phi)$$

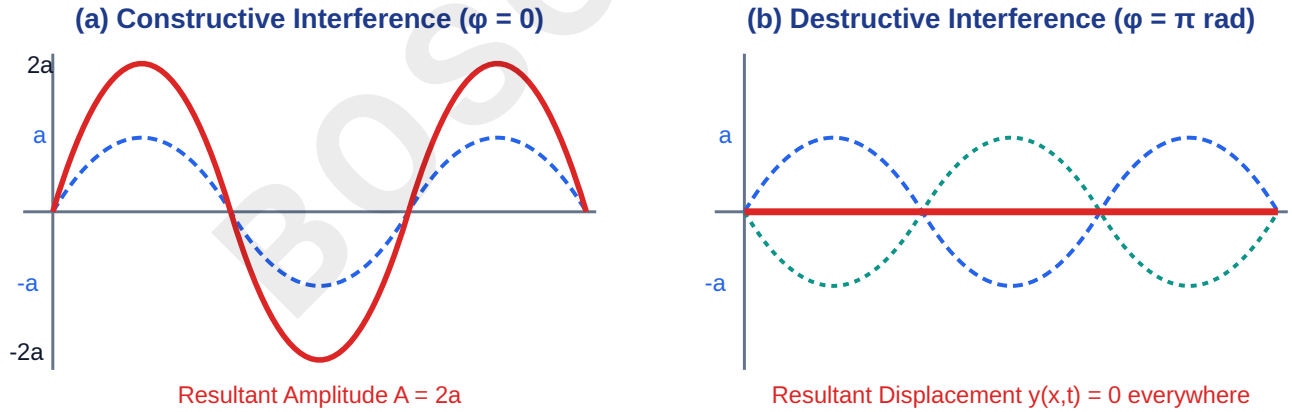
Applying the principle of superposition and the trigonometric identity $\sin A + \sin B = 2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)$:

$$y(x,t) = \left[2a\cos\left(\frac{\phi}{2}\right)\right]\sin\left(kx - \omega t + \frac{\phi}{2}\right)$$

The resultant wave is also a harmonic travelling wave of the same frequency and wavelength, with an initial phase of $\phi/2$ and a resultant amplitude $A(\phi)$ that depends on the phase difference ϕ:

$$A(\phi) = \left|2a\cos\left(\frac{\phi}{2}\right)\right|$$

- **Constructive Interference:** When $\phi = 0, 2\pi, 4\pi, \cdots = 2n\pi$ (waves are in phase), $\cos(\phi/2) = \pm 1$, and the resultant amplitude attains its maximum value $A_{\max} = 2a$:
$$y(x,t) = 2a\sin(kx - \omega t) \quad (\text{for } \phi = 0)$$
- **Destructive Interference:** When $\phi = \pi, 3\pi, 5\pi, \cdots = (2n+1)\pi$ (waves are completely out of phase), $\cos(\phi/2) = 0$, and the resultant displacement is zero everywhere at all times:
$$y(x,t) = 0$$



Tutorial Diagram 4: Superposition of two identical co-propagating waves: (a) in-phase waves add constructively to amplitude 2a; (b) out-of-phase waves cancel destructively to zero.

2.6 Reflection of Waves at Boundaries

When a travelling wave encounters an interface between two media, part of the wave is reflected and part is transmitted (refracted) into the second medium. If the incident wave is $y_i(x,t) = a\sin(kx - \omega t)$ travelling in the $+x$ direction, its reflection travels in the $-x$ direction:

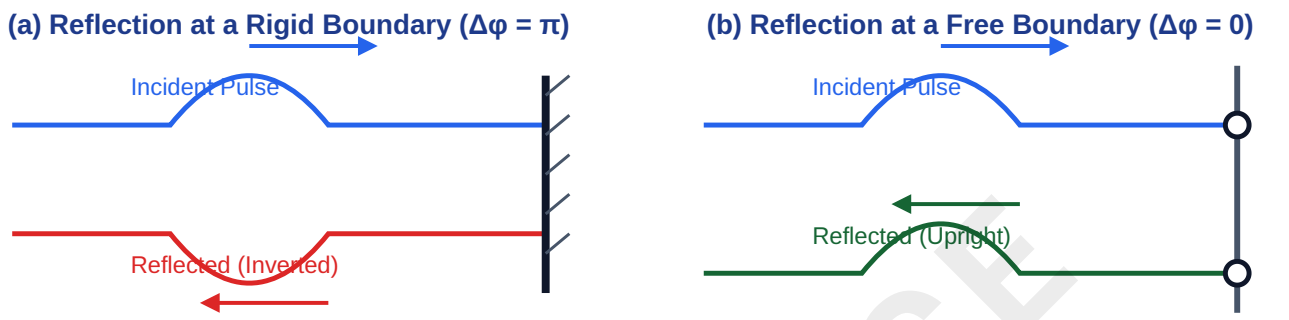
- **Reflection at a Rigid Boundary (Fixed End / Closed Pipe End):** At a rigid wall, the boundary particle cannot move, so the net displacement must be zero at all times ($y_i + y_r = 0$). Dynamically, when a crest arrives and pulls upward on the rigid wall, by Newton's third law the wall exerts an equal and opposite downward force on the string, generating an inverted reflected pulse (a trough). Thus, the wave suffers a **phase change of π radians (180°)** on reflection:

BOSONANCE

$$y_r(x,t) = a\sin(kx + \omega t + \pi) = -a\sin(kx + \omega t)$$

- **Reflection at an Open / Free Boundary (Free Ring on Rod / Open Pipe End):** When the boundary is completely free to move, there is no restoring boundary force to invert the pulse. The reflected pulse has the **same phase** ($\Delta\phi = 0$) and amplitude (assuming no energy dissipation) as the incident pulse, producing a maximum net displacement of $2a$ right at the boundary:

$$y_r(x,t) = a\sin(kx + \omega t + 0) = a\sin(kx + \omega t)$$



Tutorial Diagram 5: Reflection of a transverse pulse: (a) phase reversal of π at a rigid wall; (b) no phase change at a frictionless free ring.

2.7 Standing (Stationary) Waves and Normal Modes

In a bounded system—such as a string fixed at both ends or an air column in a pipe—waves reflect repeatedly from both boundaries. The superposition of two identical harmonic waves of equal amplitude and wavelength travelling in *opposite directions* produces a **standing wave** (or **stationary wave**). Consider:

$$y_1(x,t) = a\sin(kx - \omega t) \quad \text{and} \quad y_2(x,t) = a\sin(kx + \omega t)$$

Using $\sin(A-B) + \sin(A+B) = 2\sin A\cos B$, the resultant wave is:

$$y(x,t) = (2a\sin kx)\cos\omega t$$

(Note: If we superpose $y_i = a\sin(kx - \omega t)$ and $y_r = -a\sin(kx + \omega t)$, we obtain $y(x,t) = -(2a\sin kx)\cos\omega t$, which has the exact same position-dependent amplitude magnitude $|2a\sin kx|$.)

Key Characteristics of Standing Waves

- The spatial dependency kx and temporal dependency ωt appear *separately* as a product, not in the travelling combination $(kx \mp \omega t)$. The wave pattern does not move left or right, nor does it transport net energy across nodes.
- Every particle oscillates simple harmonically with the same angular frequency ω, but the amplitude of oscillation $A(x) = |2a\sin kx|$ varies from point to point.
- All particles within a single loop (between two consecutive nodes) vibrate **in the same phase**. Particles in adjacent loops across a node vibrate with a phase difference of π (180°).
- **Nodes (N):** Points where the amplitude is permanently zero ($\sin kx = 0$):

$$kx = n\pi \implies x = \frac{n\lambda}{2} \quad (n = 0, 1, 2, 3, \ldots)$$

- **Antinodes (A):** Points where the amplitude is maximum ($|\sin kx| = 1$, amplitude $= 2a$):

$$kx = \left(n + \frac{1}{2}\right)\pi \implies x = \left(n + \frac{1}{2}\right)\frac{\lambda}{2} \quad (n = 0, 1, 2, 3, \ldots)$$

- **Spacing Rules:**
  - Distance between any two successive nodes $= \frac{\lambda}{2}$.
  - Distance between any two successive antinodes $= \frac{\lambda}{2}$.
  - Distance between a node and the nearest adjacent antinode $= \frac{\lambda}{4}$.

BOSONANCE

Displacement Nodes vs. Pressure Antinodes in Sound Waves:

In a longitudinal sound wave, pressure variation ΔP is proportional to the volumetric strain $-\frac{\partial s}{\partial x}$ ($\Delta P = -B\frac{\partial s}{\partial x}$). At a **displacement node** (such as the closed rigid end of an organ pipe), particles cannot move ($s = 0$), but adjacent layers rush toward and away from the node from opposite sides, making the spatial gradient $\left|\frac{\partial s}{\partial x}\right|$ (compression/rarefaction) maximum. Thus:

- **Displacement Node = Pressure Antinode** (zero displacement, maximum pressure/density fluctuation).
- **Displacement Antinode = Pressure Node** (maximum displacement amplitude, minimum/zero pressure fluctuation, as at an open end exposed to constant atmospheric pressure).

2.7.1 Normal Modes of a Stretched String Fixed at Both Ends (or Open Organ Pipe)

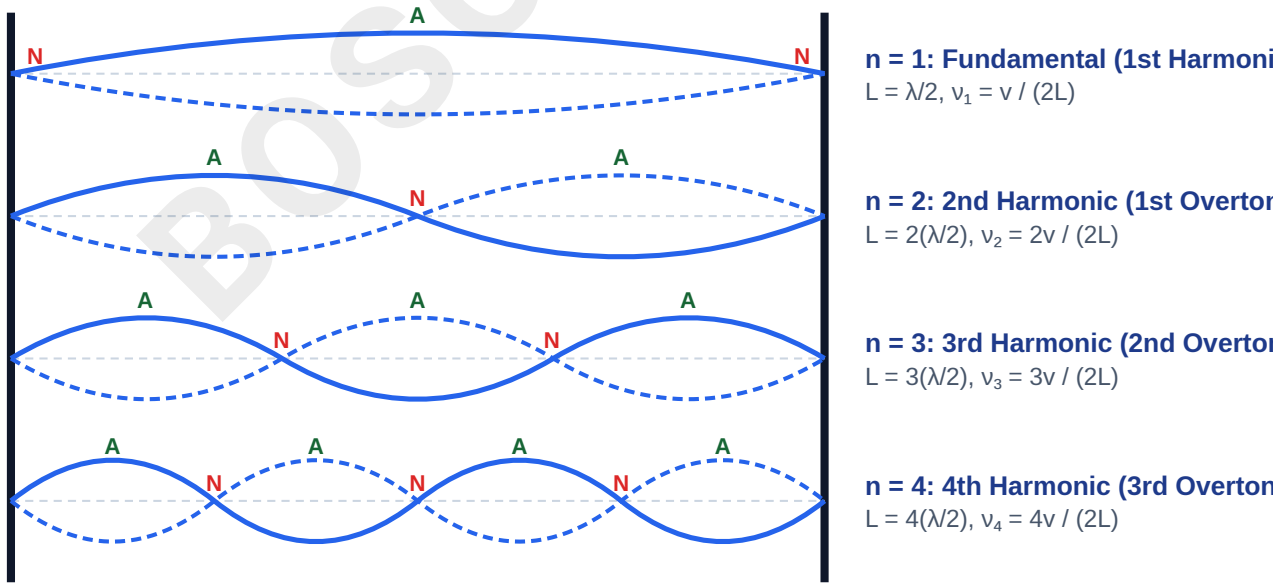
Consider a stretched string of length L rigidly clamped at both ends ($x = 0$ and $x = L$). Both ends must be displacement nodes. The condition at $x = 0$ is automatically satisfied by $\sin(kx)$. Setting a node at $x = L$ requires:

$$L = \frac{n\lambda}{2} \implies \lambda_n = \frac{2L}{n} \quad (n = 1, 2, 3, \ldots)$$

The corresponding natural frequencies (**normal modes**) are:

$$\nu_n = \frac{v}{\lambda_n} = \frac{nv}{2L} = \frac{n}{2L}\sqrt{\frac{T}{\mu}} \quad (n = 1, 2, 3, \ldots)$$

- $n = 1$: $\nu_1 = \frac{v}{2L}$ is the **fundamental mode** or **first harmonic** (1 loop, 2 nodes at ends, 1 antinode at centre).
- $n = 2$: $\nu_2 = 2\nu_1 = \frac{2v}{2L}$ is the **second harmonic** (or **first overtone**).
- $n = 3$: $\nu_3 = 3\nu_1 = \frac{3v}{2L}$ is the **third harmonic** (or **second overtone**).
- Ratio of harmonics: $\nu_1 : \nu_2 : \nu_3 : \nu_4 : \cdots = 1 : 2 : 3 : 4 : \ldots$ (**all even and odd harmonics are present**).



Tutorial Diagram 6: First four normal modes (harmonics) of a stretched string of length L fixed at both ends (N = Node, A = Antinode).

2.7.2 Normal Modes of Air Columns (Closed and Open Organ Pipes)

1. **Pipe Closed at One End and Open at the Other (e.g., Resonance Tube with Water):**
   The closed end ($x = 0$) is a displacement node (N), while the open end ($x = L$) is a displacement antinode (A). Applying the antinode condition at $x = L$:

BOSONANCE

$$L = \left(n + \frac{1}{2}\right)\frac{\lambda}{2} \implies \lambda_n = \frac{2L}{n+1/2} = \frac{4L}{2n+1} \quad (n = 0, 1, 2, 3, \ldots)$$

The normal mode frequencies of a closed organ pipe are:

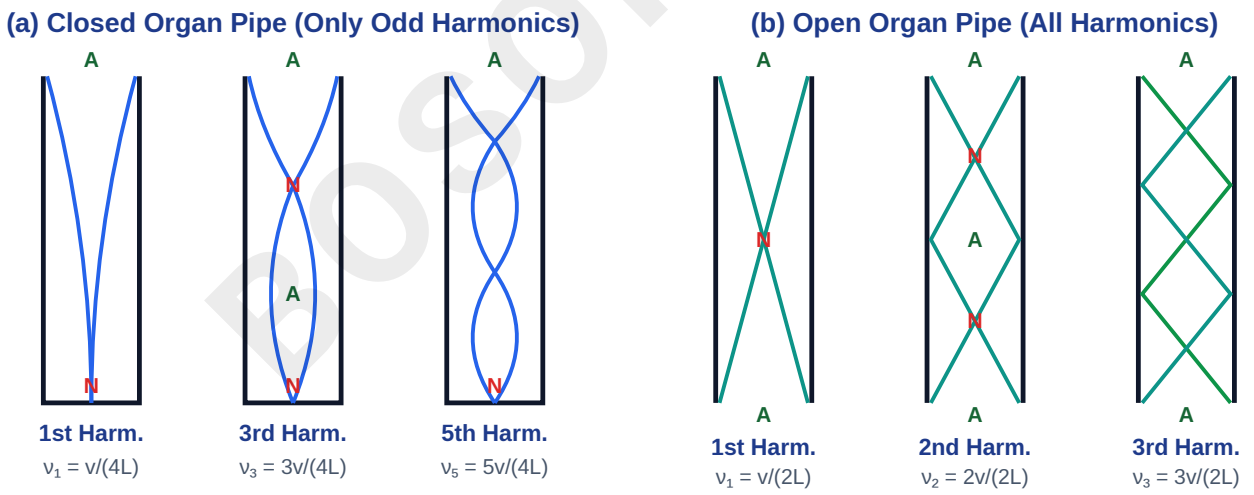
$$\nu = \left(n + \frac{1}{2}\right)\frac{v}{2L} = \frac{(2n+1)v}{4L} \quad (n = 0, 1, 2, 3, \ldots)$$

- $n = 0$: Fundamental frequency (1st harmonic) $\nu_1 = \frac{v}{4L}$ ($L = \lambda/4$).
- $n = 1$: **Third harmonic** (1st overtone) $\nu_3 = \frac{3v}{4L} = 3\nu_1$ ($L = 3\lambda/4$).
- $n = 2$: **Fifth harmonic** (2nd overtone) $\nu_5 = \frac{5v}{4L} = 5\nu_1$ ($L = 5\lambda/4$).
- **Crucial Rule:** In a pipe closed at one end, **only odd harmonics ($\nu_1 : \nu_3 : \nu_5 : \cdots = 1 : 3 : 5 : \ldots$) are possible**. Even harmonics are completely absent.

2. **Pipe Open at Both Ends:**
Both open ends ($x = 0$ and $x = L$) are displacement antinodes (A). Since the distance between consecutive antinodes is $\lambda/2$, the length L satisfies $L = \frac{n\lambda}{2}$ for $n = 1, 2, 3, \ldots$. The normal mode frequencies are:

$$\nu_n = \frac{nv}{2L} \quad (n = 1, 2, 3, \ldots)$$

- Fundamental frequency (1st harmonic): $\nu_1 = \frac{v}{2L}$ (double the fundamental frequency of a closed pipe of the same length).
- **All harmonics (both odd and even) are present:** $\nu_1 : \nu_2 : \nu_3 : \cdots = 1 : 2 : 3 : \ldots$. Because an open pipe produces richer harmonics (both even and odd), the musical quality (timbre) of an open organ pipe is richer than that of a closed pipe.



Tutorial Diagram 7: Displacement standing wave envelopes in (a) a closed organ pipe (node at closed bottom, antinode at open top) and (b) an open organ pipe (antinodes at both open ends).

2.8 Beats

While standing waves result from the superposition of waves travelling in opposite directions at the same frequency, **beats** arise from the interference in time of two harmonic sound waves of *nearly equal (close, but unequal) frequencies* ν_1 and ν_2 travelling in the same direction and heard simultaneously at a fixed location.

Fixing our location at $x = 0$ and choosing initial phase $\phi = \pi/2$ for each wave of equal amplitude a, the longitudinal displacements are:

$$s_1(t) = a\cos(\omega_1 t) \quad \text{and} \quad s_2(t) = a\cos(\omega_2 t)$$

Assuming $\omega_1 > \omega_2$ with $|\omega_1 - \omega_2| \ll \omega_1, \omega_2$, the principle of superposition and the identity $\cos A + \cos B = 2\cos\left(\frac{A-B}{2}\right)\cos\left(\frac{A+B}{2}\right)$ yield:

BOSONANCE

$$s(t) = s_1(t) + s_2(t) = \left[2a\cos\left(\frac{\omega_1 - \omega_2}{2}t\right)\right]\cos\left(\frac{\omega_1 + \omega_2}{2}t\right) = [2a\cos(\omega_b t)]\cos(\omega_a t)$$

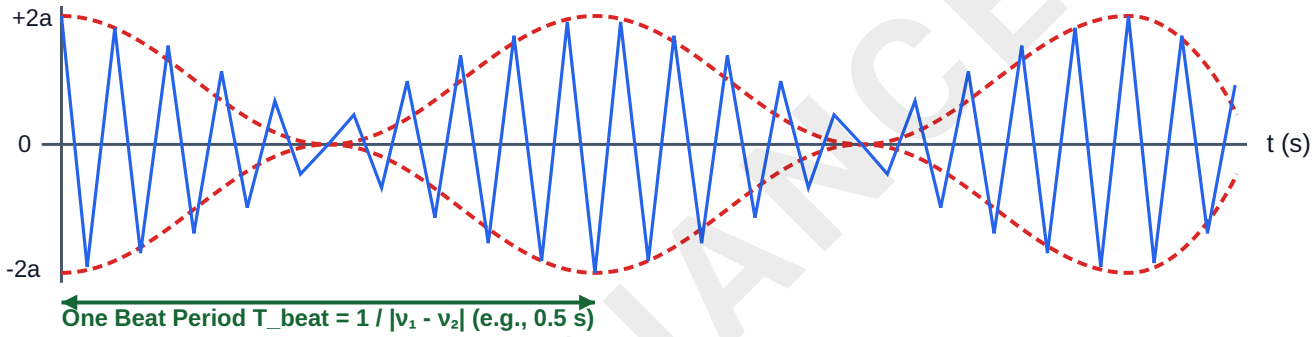
where $\omega_a = \frac{\omega_1+\omega_2}{2}$ is the average angular frequency and $\omega_b = \frac{\omega_1-\omega_2}{2}$ is the amplitude-modulation angular frequency.

Physical Interpretation of the Beat Equation

- The listener hears a note oscillating at the **average frequency** $\nu_{\text{avg}} = \frac{\nu_1+\nu_2}{2}$.
- The amplitude of this note is modulated slowly in time by the factor $|2a\cos(\omega_b t)|$. Intensity (loudness) is maximum whenever $\cos(\omega_b t) = +1$ or -1, which occurs **twice** during each full cycle of $\cos(\omega_b t)$.
- Therefore, the frequency of waxing (maximum intensity) and waning (minimum intensity) of sound—called the **beat frequency** ν_{beat}—is $2\nu_b$:

$$\nu_{\text{beat}} = |\nu_1 - \nu_2|$$

Superposition of Two Close Frequencies (e.g., 11 Hz and 9 Hz) Producing Beats (2 Hz)



Tutorial Diagram 8: Beats waveform showing rapid carrier oscillation at average frequency $(v_1 + v_2)/2$ modulated by a slow envelope of beat frequency $|v_1 - v_2|$.

Real-World Physics Application — Acoustic Architecture of Musical Pillars:

At historical South Indian temples such as the 7th-century Nellaiappar temple in Tamil Nadu, Hampi, Kanyakumari, and Thiruvananthapuram, clusters of stone pillars carved from a single rock produce distinct musical notes (*Sa, Re, Ga, Ma, Pa, Dha, Ni*) when gently tapped. The natural frequencies of these pillars depend on the **elasticity (Young's modulus Y) of the stone, its mass density (ρ), and its geometric dimensions**. These pillars are classified into three acoustic categories: *Shruti Pillars* (producing basic notes/swaras), *Gana Thoongal* (generating raga tunes), and *Laya Thoongal* (producing rhythmic taal/beats).

BOSONANCE

3. Master Formula & Physical Quantities Compendium

Summary of Physical Quantities, Dimensions, SI Units, and Defining Relations

Physical Quantity	Symbol	Dimensions	SI Unit	Defining Relation / Remarks
Wave Displacement	$y(x,t)$ or $s(x,t)$	$[\mathrm{L}]$	m	$y(x,t) = a\sin(kx - \omega t + \phi)$
Wavelength	λ	$[\mathrm{L}]$	m	Distance between two consecutive points in the same phase.
Propagation Constant (Angular Wave Number)	k	$[\mathrm{L}^{-1}]$	$\mathrm{rad\,m^{-1}}$ or $\mathrm{m^{-1}}$	$k = \frac{2\pi}{\lambda}$; phase change per unit path difference: $\Delta\phi = k\Delta x$.
Angular Frequency	ω	$[\mathrm{T}^{-1}]$	$\mathrm{rad\,s^{-1}}$	$\omega = \frac{2\pi}{T} = 2\pi\nu$; phase change in time Δt: $\Delta\phi = \omega\Delta t$.
Wave Speed	v	$[\mathrm{LT}^{-1}]$	$\mathrm{m\,s^{-1}}$	$v = \frac{\omega}{k} = \frac{\lambda}{T} = \nu\lambda$
Linear Mass Density	μ	$[\mathrm{ML}^{-1}]$	$\mathrm{kg\,m^{-1}}$	$\mu = \frac{m}{L} = \rho A$; $v_{\text{string}} = \sqrt{\frac{T}{\mu}}$
Bulk Modulus	B	$[\mathrm{ML}^{-1}\mathrm{T}^{-2}]$	Pa ($\mathrm{N\,m^{-2}}$)	$B = -\frac{\Delta P}{\Delta V/V}$; $v_{\text{fluid}} = \sqrt{\frac{B}{\rho}}$; $B_{\text{ad}} = \gamma P$ in gases.
Beat Frequency	ν_{beat}	$[\mathrm{T}^{-1}]$	Hz ($\mathrm{s^{-1}}$)	$\nu_{\text{beat}} = \lvert\nu_1 - \nu_2\rvert$

4. Problem-Solving Procedures & Conceptual Connections

4.1 How to Test Whether a Function $y = f(x,t)$ Represents a Travelling Wave

1. **Combination Rule:** In a one-dimensional undistorted travelling wave, x and t must appear strictly in the linear combination $(x - vt)$ or $(x + vt)$ (or equivalently $kx \mp \omega t$).
2. **Finiteness and Continuity Rule (Crucial Converse Check!):** Having the form $f(x \pm vt)$ is a *necessary* condition, but **not sufficient**! For a function $y = f(x \pm vt)$ to physically represent a realistic travelling wave, the displacement y must remain **finite, continuous, and bounded** for all values of $x \in (-\infty, +\infty)$ and $t \in (-\infty, +\infty)$.
   - $y = (x - vt)^2$: Diverges to ∞ as $x \to \infty$. **Not a travelling wave.**
   - $y = \log[(x + vt)/x_0]$: Diverges at $x + vt \to 0$ and ∞. **Not a travelling wave.**
   - $y = \frac{1}{x+vt}$: Blows up to ∞ when $x + vt = 0$. **Not a travelling wave.**
   - $y = 2\sqrt{x - vt}$: Imaginary for $x < vt$ and unbounded for $x \to \infty$. **Not a travelling wave.**

4.2 Decision Tree for Beat Problems (Tension & Loading Adjustments)

Suppose Source A has known frequency ν_A, Source B has unknown frequency ν_B, and initially they produce b_1 beats per second. Thus, ν_B can only be one of two values: $\nu_B = \nu_A + b_1$ or $\nu_B = \nu_A - b_1$.

- **Case 1: Tension in String B is Increased (ν_B increases because $\nu \propto \sqrt{T}$):**
  - If ν_B was originally $\nu_A + b_1$ (higher than ν_A), increasing ν_B moves it further away from ν_A, so the beat frequency **increases**.
  - If ν_B was originally $\nu_A - b_1$ (lower than ν_A), slightly increasing ν_B brings it closer to ν_A, so the beat frequency **decreases**.

- **Case 2: Tension in String A is Reduced (ν_A decreases):**
  - If $\nu_B = \nu_A - b_1$ (so $\nu_A > \nu_B$), slightly decreasing ν_A brings ν_A closer to ν_B, so the beat frequency **decreases**.
  - If $\nu_B = \nu_A + b_1$ (so $\nu_A < \nu_B$), decreasing ν_A moves it further below ν_B, so the beat frequency **increases**.

4.3 Key Conceptual Answers (Why & How)

- **How bats navigate without eyes (Echolocation):** Bats emit high-frequency, short-wavelength ultrasonic waves. Because ultrasonic wavelengths are very small, they undergo minimal diffraction and reflect sharply off obstacles and prey. By detecting the time delay, direction, and intensity of the reflected echoes, bats ascertain the distance, direction, nature, and size of obstacles.
- **Why a violin note and a sitar note of the same fundamental frequency sound distinct:** Musical instruments do not vibrate purely in their fundamental mode; they simultaneously excite multiple higher harmonics (overtones) with different relative intensities depending on instrument geometry and how the string is bowed or plucked. While their fundamental frequency (pitch) is identical, the relative amplitudes and number of overtones—known as **timbre (quality)**—differ, allowing the human ear to distinguish them.

BOSONANCE

- **Why a pulse gets distorted in a dispersive medium:** Any non-sinusoidal pulse is a Fourier superposition of many harmonic waves of different frequencies/wavelengths. In a *dispersive medium*, the wave speed v depends on frequency. As the pulse propagates, its different frequency components travel at different speeds and get out of phase, distorting the shape of the pulse.

5. Comprehensive Worked Examples

Worked Example 1: Classification of Wave Motions

Problem: State in each of the following cases whether the wave motion is transverse, longitudinal, or a combination of both:

a. Motion of a kink in a longitudinal spring produced by displacing one end of the spring sideways.

b. Waves produced in a cylinder containing a liquid by moving its piston back and forth.

c. Waves produced by a motorboat sailing in water.

d. Ultrasonic waves in air produced by a vibrating quartz crystal.

Solution:

a. **Transverse and longitudinal (combination of both):** Displacing the spring sideways introduces transverse oscillations normal to the spring's length while simultaneously stretching the coils longitudinally along the spring.

b. **Longitudinal:** Moving the piston back and forth compresses and expands the liquid along the axis of the cylinder; furthermore, bulk liquids cannot sustain shear stress.

c. **Transverse and longitudinal (combination of both):** Surface water waves created by a motorboat involve water particles moving both vertically (up and down) and horizontally (back and forth).

d. **Longitudinal:** Air is a gas and possesses only volume (bulk) elasticity, not shear rigidity; hence ultrasonic sound waves in air are purely longitudinal.

Worked Example 2: Extracting Parameters from a Progressive Wave Equation

Problem: A wave travelling along a string is described by:

$$y(x,t) = 0.005 \sin(80.0x - 3.0t)$$

in which the numerical constants are in SI units ($0.005\,\text{m}$, $80.0\,\text{rad}\,\text{m}^{-1}$, and $3.0\,\text{rad}\,\text{s}^{-1}$). Calculate (a) the amplitude, (b) the wavelength, (c) the period and frequency of the wave, and (d) the displacement y of the wave at a distance $x = 30.0\,\text{cm}$ and time $t = 20\,\text{s}$.

Solution:

Comparing the given equation with the standard progressive wave equation $y(x,t) = a\sin(kx - \omega t)$:

a. **Amplitude:** $a = 0.005\,\text{m} = 5\,\text{mm}$.

b. **Wavelength:** Given angular wave number $k = 80.0\,\text{rad}\,\text{m}^{-1}$:

$$\lambda = \frac{2\pi}{k} = \frac{2\pi}{80.0\,\text{m}^{-1}} = 0.0785\,\text{m} = 7.85\,\text{cm}$$

c. **Time Period and Frequency:** Given angular frequency $\omega = 3.0\,\text{rad}\,\text{s}^{-1}$:

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{3.0\,\text{s}^{-1}} = 2.09\,\text{s}$$

$$\nu = \frac{1}{T} = \frac{1}{2.09\,\text{s}} = 0.48\,\text{Hz}$$

d. **Displacement at $x = 30.0\,\text{cm} = 0.30\,\text{m}$ and $t = 20\,\text{s}$:**

$y(0.30, 20) = (0.005\,\text{m})\sin(80.0 \times 0.30 - 3.0 \times 20) = (0.005\,\text{m})\sin(24 - 60) = (0.005\,\text{m})\sin(-36\,\text{rad})$

Adding $12\pi \approx 37.699\,\text{rad}$ to bring the angle into $[0, 2\pi]$:

$y = (0.005\,\text{m})\sin(-36 + 12\pi) = (0.005\,\text{m})\sin(1.699\,\text{rad}) \approx (0.005\,\text{m})\sin(97^\circ) \approx 0.00496\,\text{m} \approx 5\,\text{mm}$

BOSONANCE

Worked Example 3: Wave Speed and Travel Time on Stretched Strings

Problem:

a. A steel wire $0.72\,\text{m}$ long has a mass of $5.0 \times 10^{-3}\,\text{kg}$. If the wire is under a tension of $60\,\text{N}$, what is the speed of transverse waves on the wire?

b. A string of mass $2.50\,\text{kg}$ is under a tension of $200\,\text{N}$. The length of the stretched string is $20.0\,\text{m}$. If a transverse jerk is struck at one end of the string, how long does the disturbance take to reach the other end?

Solution:

a. Linear mass density of the steel wire:

$$\mu = \frac{m}{L} = \frac{5.0 \times 10^{-3}\,\text{kg}}{0.72\,\text{m}} = 6.94 \times 10^{-3}\,\text{kg m}^{-1}$$

Given tension $T = 60\,\text{N}$, the speed of transverse waves is:

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{60\,\text{N}}{6.94 \times 10^{-3}\,\text{kg m}^{-1}}} \approx 93\,\text{m s}^{-1}$$

b. For the second string, linear mass density is:

$$\mu = \frac{m}{L} = \frac{2.50\,\text{kg}}{20.0\,\text{m}} = 0.125\,\text{kg m}^{-1}$$

Speed of the transverse wave on the string:

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{200\,\text{N}}{0.125\,\text{kg m}^{-1}}} = \sqrt{1600} = 40.0\,\text{m s}^{-1}$$

Time taken by the disturbance to travel the length $L = 20.0\,\text{m}$:

$$t = \frac{L}{v} = \frac{20.0\,\text{m}}{40.0\,\text{m s}^{-1}} = 0.50\,\text{s}$$

Worked Example 4: Free Fall and Sound Propagation Delay

Problem: A stone dropped from the top of a tower of height $300\,\text{m}$ splashes into the water of a pond near the base of the tower. When is the splash heard at the top, given that the speed of sound in air is $340\,\text{m s}^{-1}$? (Take $g = 9.8\,\text{m s}^{-2}$).

Solution:

The total time t_{total} is the sum of two intervals: (1) time t_1 taken by the stone to fall freely through $h = 300\,\text{m}$ from rest, and (2) time t_2 taken by the sound of the splash to travel upward through $h = 300\,\text{m}$ at constant speed $v = 340\,\text{m s}^{-1}$.

Step 1 (Free-fall time t_1): Using $h = \frac{1}{2}gt_1^2$:

$$t_1 = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2 \times 300\,\text{m}}{9.8\,\text{m s}^{-2}}} = \sqrt{61.224} \approx 7.82\,\text{s}$$

Step 2 (Sound travel time t_2):

$$t_2 = \frac{h}{v} = \frac{300\,\text{m}}{340\,\text{m s}^{-1}} \approx 0.88\,\text{s}$$

Step 3 (Total time):

$$t_{\text{total}} = t_1 + t_2 = 7.82\,\text{s} + 0.88\,\text{s} = 8.70\,\text{s}$$

BOSONANCE

Worked Example 5: Phase Difference vs. Path Difference in a Travelling Wave

Problem: For the travelling harmonic wave

$$y(x,t) = 2.0\cos[2\pi(10t - 0.0080x + 0.35)]$$

where x and y are in cm and t in s, calculate the phase difference between the oscillatory motion of two points separated by a distance of: (a) $4\,\mathrm{m}$, (b) $0.5\,\mathrm{m}$, (c) $\lambda/2$, and (d) $3\lambda/4$.

Solution:

Rewriting the wave equation in standard form $y(x,t) = 2.0\cos(20\pi t - 0.016\pi x + 0.70\pi)$, we identify the angular wave number k:

$$k = 2\pi \times 0.0080\,\mathrm{cm}^{-1} = 0.016\pi\,\mathrm{cm}^{-1} = 1.6\pi\,\mathrm{m}^{-1}$$

At any given instant t, the magnitude of phase difference $\Delta\phi$ between two points separated by path difference Δx is $\Delta\phi = k\Delta x = \frac{2\pi}{\lambda}\Delta x$:

a. For $\Delta x = 4\,\mathrm{m}$: $\Delta\phi = (1.6\pi\,\mathrm{m}^{-1})(4\,\mathrm{m}) = 6.4\pi\,\mathrm{rad}$.

b. For $\Delta x = 0.5\,\mathrm{m}$: $\Delta\phi = (1.6\pi\,\mathrm{m}^{-1})(0.5\,\mathrm{m}) = 0.8\pi\,\mathrm{rad}$.

c. For $\Delta x = \lambda/2$: $\Delta\phi = \frac{2\pi}{\lambda}\left(\frac{\lambda}{2}\right) = \pi\,\mathrm{rad}$.

d. For $\Delta x = 3\lambda/4$: $\Delta\phi = \frac{2\pi}{\lambda}\left(\frac{3\lambda}{4}\right) = 1.5\pi\,\mathrm{rad} = \frac{3\pi}{2}\,\mathrm{rad}$.

BOSONANCE

Worked Example 6: Complete Analysis of a Standing Wave on a Clamped String

Problem: The transverse displacement of a string clamped at both its ends is given by:

$$y(x,t) = 0.06 \sin\left(\frac{2\pi}{3}x\right)\cos(120\pi t)$$

where x and y are in m and t in s. The length of the string is $1.5\,\mathrm{m}$ and its mass is $3.0 \times 10^{-2}\,\mathrm{kg}$.

a. Does the function represent a travelling wave or a stationary wave?

b. Interpret the wave as a superposition of two waves travelling in opposite directions. What are the wavelength, frequency, and speed of each wave?

c. Determine the tension in the string.

d. What is the amplitude of oscillation of a point at $x = 0.375\,\mathrm{m}$ away from one end?

Solution:

a. Because the spatial factor $\sin\left(\frac{2\pi}{3}x\right)$ and temporal factor $\cos(120\pi t)$ appear as separate multiplied functions rather than in the form $(kx \mp \omega t)$, the function represents a **stationary (standing) wave**.

b. Comparing with $y(x,t) = 2a\sin(kx)\cos(\omega t) = a\sin(kx-\omega t) + a\sin(kx+\omega t)$:

- Amplitude of each component travelling wave: $a = \frac{0.06\,\mathrm{m}}{2} = 0.03\,\mathrm{m}$.
- Wave number $k = \frac{2\pi}{3}\,\mathrm{m}^{-1} \implies$ Wavelength $\lambda = \frac{2\pi}{k} = 3.0\,\mathrm{m}$.
- Angular frequency $\omega = 120\pi\,\mathrm{rad\,s}^{-1} \implies$ Frequency $\nu = \frac{\omega}{2\pi} = \frac{120\pi}{2\pi} = 60\,\mathrm{Hz}$.
- Speed of each component wave: $v = \nu\lambda = (60\,\mathrm{Hz})(3.0\,\mathrm{m}) = 180\,\mathrm{m\,s}^{-1}$.

c. Linear mass density of the string:

$$\mu = \frac{m}{L} = \frac{3.0 \times 10^{-2}\,\mathrm{kg}}{1.5\,\mathrm{m}} = 2.0 \times 10^{-2}\,\mathrm{kg\,m}^{-1}$$

Using $v = \sqrt{T/\mu}$, the tension T is:

$$T = \mu v^2 = (2.0 \times 10^{-2}\,\mathrm{kg\,m}^{-1})(180\,\mathrm{m\,s}^{-1})^2 = 0.02 \times 32400 = 648\,\mathrm{N}$$

d. In a stationary wave, all points oscillate with the same frequency ($\nu = 60\,\mathrm{Hz}$) and all points within the single loop (since $L = 1.5\,\mathrm{m} = \lambda/2$) have the same phase, but different amplitudes $A(x) = \left|0.06\sin\left(\frac{2\pi}{3}x\right)\right|$. At $x = 0.375\,\mathrm{m}$:

$$A(0.375) = 0.06\sin\left(\frac{2\pi}{3} \times 0.375\right) = 0.06\sin\left(\frac{\pi}{4}\right) = 0.06 \times 0.707 = 0.0424\,\mathrm{m} = 4.24\,\mathrm{cm}$$

BOSONANCE

Worked Example 7: Resonance in Open vs. Closed Organ Pipes

Problem: A pipe $30.0\,\text{cm}$ long is open at both ends. Which harmonic mode of the pipe resonates with a $1.1\,\text{kHz}$ source? Will resonance with the same source be observed if one end of the pipe is closed? Take the speed of sound in air as $330\,\text{m s}^{-1}$.

Solution:

Part 1 (Pipe open at both ends): Length $L = 30.0\,\text{cm} = 0.30\,\text{m}$. The nth harmonic frequency of an open pipe is:

$$\nu_n = \frac{nv}{2L} = \frac{n \times 330\,\text{m s}^{-1}}{2 \times 0.30\,\text{m}} = 550n\,\text{Hz} \quad (n = 1, 2, 3, \ldots)$$

Setting $\nu_n = 1.1\,\text{kHz} = 1100\,\text{Hz}$:

$$550n = 1100 \implies n = 2$$

Thus, the open pipe resonates in its **second harmonic** mode.

Part 2 (One end of the pipe closed): For a pipe of length $L = 0.30\,\text{m}$ closed at one end, the fundamental frequency is:

$$\nu_1' = \frac{v}{4L} = \frac{330\,\text{m s}^{-1}}{4 \times 0.30\,\text{m}} = 275\,\text{Hz}$$

Because a closed pipe supports *only odd harmonics*, its allowed resonant frequencies are $\nu_1' = 275\,\text{Hz}$, $\nu_3' = 3 \times 275 = 825\,\text{Hz}$, $\nu_5' = 5 \times 275 = 1375\,\text{Hz}$, etc. Since $1100\,\text{Hz} = 4 \times 275\,\text{Hz}$ (an even multiple, which is forbidden), **no resonance will be observed** when one end is closed.

Worked Example 8: Resonance Tube Speed of Sound & Clamped Steel Rod

Problem:

a. A metre-long tube open at one end, with a movable piston at the other end, shows resonance with a fixed frequency tuning fork of $340\,\text{Hz}$ when the tube length is $25.5\,\text{cm}$ or $79.3\,\text{cm}$. Estimate the speed of sound in air at the temperature of the experiment (neglect edge effects).

b. A steel rod $100\,\text{cm}$ long is clamped at its middle. The fundamental frequency of longitudinal vibrations of the rod is $2.53\,\text{kHz}$. What is the speed of sound in steel?

Solution:

a. A tube with a movable piston acts as a pipe closed at one end. Its resonances occur at odd multiples of $\lambda/4$. Since $L_1 = 25.5\,\text{cm}$ and $L_2 = 79.3\,\text{cm}$ (which is approximately $3L_1$), these correspond to the fundamental mode ($L_1 = \lambda/4$) and third harmonic ($L_2 = 3\lambda/4$):

$$L_2 - L_1 = \frac{3\lambda}{4} - \frac{\lambda}{4} = \frac{\lambda}{2} \implies \lambda = 2(L_2 - L_1) = 2(79.3 - 25.5)\,\text{cm} = 107.6\,\text{cm} = 1.076\,\text{m}$$

(Alternatively, directly from $L_1 = \lambda/4 \implies \lambda = 4 \times 0.255\,\text{m} = 1.02\,\text{m}$, giving $v = 340 \times 1.02 = 346.8\,\text{m s}^{-1}$; using the difference $L_2 - L_1$ gives $v = 340 \times 1.076 \approx 365.8\,\text{m s}^{-1}$).

b. When the steel rod of length $L = 100\,\text{cm} = 1.00\,\text{m}$ is clamped at its midpoint, the midpoint is a displacement **node** (N) and both free ends are displacement **antinodes** (A). In the fundamental mode, the distance from one free end (A) to the other free end (A) with a single node in between is:

$$L = \frac{\lambda}{2} \implies \lambda = 2L = 2 \times 1.00\,\text{m} = 2.00\,\text{m}$$

Given fundamental frequency $\nu = 2.53\,\text{kHz} = 2530\,\text{Hz}$, the speed of longitudinal sound waves in steel is:

$$v = \nu\lambda = (2530\,\text{Hz})(2.00\,\text{m}) = 5060\,\text{m s}^{-1} = 5.06\,\text{km s}^{-1}$$

BOSONANCE

Worked Example 9: Beats and String Tension Adjustment

Problem:

a. Two sitar strings A and B playing the note 'Dha' are slightly out of tune and produce beats of frequency $5\ \text{Hz}$. The tension of string B is slightly increased and the beat frequency is found to decrease to $3\ \text{Hz}$. What is the original frequency of B if the frequency of A is $427\ \text{Hz}$?

b. Two sitar strings A and B playing the note 'Ga' produce beats of frequency $6\ \text{Hz}$. The tension in string A is slightly reduced and the beat frequency reduces to $3\ \text{Hz}$. If the original frequency of A is $324\ \text{Hz}$, what is the frequency of B?

Solution:

a. Given $\nu_A = 427\ \text{Hz}$ and initial beat frequency $|\nu_A - \nu_B| = 5\ \text{Hz}$, so originally $\nu_B = 432\ \text{Hz}$ or $422\ \text{Hz}$. Increasing the tension of string B increases ν_B. If ν_B were $432\ \text{Hz}$ ($\nu_B > \nu_A$), increasing ν_B would increase the difference $|\nu_B - \nu_A|$ above $5\ \text{Hz}$. Since the beat frequency *decreases* to $3\ \text{Hz}$, we must have $\nu_B < \nu_A$. Therefore:

$$\nu_B = \nu_A - 5\ \text{Hz} = 427 - 5 = 422\ \text{Hz}$$

b. Here $\nu_A = 324\ \text{Hz}$ and $|\nu_A - \nu_B| = 6\ \text{Hz}$, so $\nu_B = 330\ \text{Hz}$ or $318\ \text{Hz}$. Reducing the tension in string A *decreases* ν_A. Since reducing ν_A reduces the gap $|\nu_A - \nu_B|$ from $6\ \text{Hz}$ to $3\ \text{Hz}$, ν_A must originally have been higher than ν_B ($\nu_A > \nu_B$). Therefore:

$$\nu_B = \nu_A - 6\ \text{Hz} = 324 - 6 = 318\ \text{Hz}$$

6. NEET UG Exam-Style Practice Questions

Note: All questions below are original practice items designed strictly in accordance with the NEET UG Physics syllabus and single-choice objective format.

Practice Question 1 (Wave Equation & Propagation Velocity)

A transverse harmonic wave on a long string is represented by the equation:

$$y(x,t) = 3.0\sin\left(36t + 0.018x + \frac{\pi}{4}\right)$$

where x and y are in cm and t is in s. Taking the positive x-direction from left to right, which of the following gives the correct speed of propagation, direction of travel, and least distance between two successive crests?

(A) $20\ \text{m s}^{-1}$, left to right, $3.49\ \text{m}$

(B) $20\ \text{m s}^{-1}$, right to left, $3.49\ \text{m}$

(C) $200\ \text{m s}^{-1}$, right to left, $0.35\ \text{m}$

(D) $0.65\ \text{m s}^{-1}$, left to right, $3.49\ \text{m}$

Correct Answer: (B)

Worked Solution:

Comparing $y(x,t) = 3.0\sin(36t + 0.018x + \pi/4)$ with $y(x,t) = a\sin(\omega t + kx + \phi)$:

- Because ωt and kx have the **same positive sign**, the wave is a travelling wave propagating in the negative x-direction, i.e., from **right to left**.
- Here $\omega = 36\ \text{rad s}^{-1}$ and $k = 0.018\ \text{rad cm}^{-1}$. The wave speed is:

$$v = \frac{\omega}{k} = \frac{36\ \text{s}^{-1}}{0.018\ \text{cm}^{-1}} = 2000\ \text{cm s}^{-1} = 20\ \text{m s}^{-1}$$

- The least distance between two successive crests is the wavelength λ:

$$\lambda = \frac{2\pi}{k} = \frac{2 \times 3.1416}{0.018\ \text{cm}^{-1}} \approx 349\ \text{cm} = 3.49\ \text{m}$$

BOSONANCE

Practice Question 2 (Wave Speed on a Stretched Wire)

A uniform steel wire has a length of $12.0\,\mathrm{m}$ and a mass of $2.10\,\mathrm{kg}$. What should be the tension in the wire so that the speed of a transverse wave on the wire equals the speed of sound in dry air at $20\,^{\circ}\mathrm{C}$ ($343\,\mathrm{m\,s^{-1}}$)?

(A) $2.06 \times 10^{4}\,\mathrm{N}$

(B) $1.03 \times 10^{4}\,\mathrm{N}$

(C) $6.00 \times 10^{3}\,\mathrm{N}$

(D) $4.12 \times 10^{4}\,\mathrm{N}$

Correct Answer: (A)

Worked Solution:

Linear mass density of the wire:

$$\mu = \frac{m}{L} = \frac{2.10\,\mathrm{kg}}{12.0\,\mathrm{m}} = 0.175\,\mathrm{kg\,m^{-1}}$$

From $v = \sqrt{\frac{T}{\mu}}$, the required tension T for $v = 343\,\mathrm{m\,s^{-1}}$ is:

$$T = \mu v^{2} = (0.175\,\mathrm{kg\,m^{-1}}) \times (343\,\mathrm{m\,s^{-1}})^{2} = 0.175 \times 117649 \approx 2.06 \times 10^{4}\,\mathrm{N}$$

Practice Question 3 (Reflected vs. Transmitted Ultrasonic Waves)

A bat flying in air emits ultrasonic sound of frequency $1000\,\mathrm{kHz}$. If this sound meets a flat water surface, what are the wavelengths of (i) the reflected sound in air, and (ii) the transmitted sound in water? (Given: speed of sound in air $= 340\,\mathrm{m\,s^{-1}}$, speed of sound in water $= 1486\,\mathrm{m\,s^{-1}}$).

(A) (i) $3.4 \times 10^{-4}\,\mathrm{m}$, (ii) $3.4 \times 10^{-4}\,\mathrm{m}$

(B) (i) $3.4 \times 10^{-4}\,\mathrm{m}$, (ii) $1.49 \times 10^{-3}\,\mathrm{m}$

(C) (i) $1.49 \times 10^{-3}\,\mathrm{m}$, (ii) $3.4 \times 10^{-4}\,\mathrm{m}$

(D) (i) $3.4 \times 10^{-3}\,\mathrm{m}$, (ii) $1.49 \times 10^{-2}\,\mathrm{m}$

Correct Answer: (B)

Worked Solution:

Frequency $\nu = 1000\,\mathrm{kHz} = 1.0 \times 10^{6}\,\mathrm{Hz}$ is determined solely by the source and remains unchanged upon reflection and transmission.

- (i) Reflected sound travels back in air ($v_{\mathrm{air}} = 340\,\mathrm{m\,s^{-1}}$):

$$\lambda_{\mathrm{ref}} = \frac{v_{\mathrm{air}}}{\nu} = \frac{340\,\mathrm{m\,s^{-1}}}{1.0 \times 10^{6}\,\mathrm{Hz}} = 3.4 \times 10^{-4}\,\mathrm{m}$$

- (ii) Transmitted sound travels in water ($v_{\mathrm{water}} = 1486\,\mathrm{m\,s^{-1}}$):

$$\lambda_{\mathrm{trans}} = \frac{v_{\mathrm{water}}}{\nu} = \frac{1486\,\mathrm{m\,s^{-1}}}{1.0 \times 10^{6}\,\mathrm{Hz}} = 1.486 \times 10^{-3}\,\mathrm{m} \approx 1.49 \times 10^{-3}\,\mathrm{m}$$

BOSONANCE

Practice Question 4 (Medical Ultrasonic Scanner Wavelength)

A hospital uses an ultrasonic scanner operating at a frequency of $4.2\,\mathrm{MHz}$ to locate tumours in a biological tissue. If the speed of sound in the tissue is $1.7\,\mathrm{km\,s^{-1}}$, the wavelength of sound in the tissue is approximately:

(A) $4.05 \times 10^{-4}\,\mathrm{m}$

(B) $2.47 \times 10^{-3}\,\mathrm{m}$

(C) $4.05 \times 10^{-6}\,\mathrm{m}$

(D) $7.14 \times 10^{-3}\,\mathrm{m}$

Correct Answer: (A)

Worked Solution:

Given wave speed $v = 1.7\,\mathrm{km\,s^{-1}} = 1.7 \times 10^{3}\,\mathrm{m\,s^{-1}}$ and operating frequency $\nu = 4.2\,\mathrm{MHz} = 4.2 \times 10^{6}\,\mathrm{Hz}$:

$$\lambda = \frac{v}{\nu} = \frac{1.7 \times 10^{3}\,\mathrm{m\,s^{-1}}}{4.2 \times 10^{6}\,\mathrm{Hz}} = 4.048 \times 10^{-4}\,\mathrm{m} \approx 4.05 \times 10^{-4}\,\mathrm{m}$$

Practice Question 5 (Identifying Travelling vs. Stationary Wave Functions)

Consider the following four mathematical functions of position x and time t:

1. $y_1 = 2\cos(3x)\sin(10t)$
2. $y_2 = 2\sqrt{x - vt}$
3. $y_3 = 3\sin(5x - 0.5t) + 4\cos(5x - 0.5t)$
4. $y_4 = \cos(x)\sin(t) + \cos(2x)\sin(2t)$

Which of the following correctly classifies these functions?

(A) y_1, y_4: stationary waves; y_3: travelling wave; y_2: neither

(B) y_1: stationary wave; y_2, y_3: travelling waves; y_4: neither

(C) y_1, y_3: travelling waves; y_4: stationary wave; y_2: neither

(D) y_1: stationary wave; y_3: travelling wave; y_2, y_4: neither

Correct Answer: (A)

Worked Solution:

- $y_1 = 2\cos(3x)\sin(10t)$ has separated spatial and temporal harmonic factors, representing a **stationary wave**.
- $y_2 = 2\sqrt{x - vt}$ is not bounded for all x, t and becomes imaginary when $x < vt$; hence it represents **neither**.
- $y_3 = 3\sin(5x - 0.5t) + 4\cos(5x - 0.5t) = 5\sin(5x - 0.5t + \phi)$ is a bounded function of $(5x - 0.5t)$, representing a harmonic **travelling wave** of amplitude 5.
- $y_4 = \cos(x)\sin(t) + \cos(2x)\sin(2t)$ is a superposition of two stationary wave modes, hence representing a **stationary wave**.

BOSONANCE

Practice Question 6 (Fundamental Vibration of a Stretched Wire)

A wire stretched between two rigid supports vibrates in its fundamental mode with a frequency of $45\,\text{Hz}$. The mass of the wire is $3.5 \times 10^{-2}\,\text{kg}$ and its linear mass density is $4.0 \times 10^{-2}\,\text{kg m}^{-1}$. What are (i) the speed of a transverse wave on the string, and (ii) the tension in the string?

(A) (i) $78.75\,\text{m s}^{-1}$, (ii) $248\,\text{N}$

(B) (i) $39.38\,\text{m s}^{-1}$, (ii) $62\,\text{N}$

(C) (i) $78.75\,\text{m s}^{-1}$, (ii) $124\,\text{N}$

(D) (i) $157.5\,\text{m s}^{-1}$, (ii) $248\,\text{N}$

Correct Answer: (A)

Worked Solution:

Given mass $m = 3.5 \times 10^{-2}\,\text{kg}$ and linear mass density $\mu = \frac{m}{L} = 4.0 \times 10^{-2}\,\text{kg m}^{-1}$, the length L of the wire is:

$$L = \frac{m}{\mu} = \frac{3.5 \times 10^{-2}\,\text{kg}}{4.0 \times 10^{-2}\,\text{kg m}^{-1}} = 0.875\,\text{m}$$

- (i) In the fundamental mode, $\nu_1 = \frac{v}{2L}$, so the wave speed is:
$$v = 2L\nu_1 = 2 \times (0.875\,\text{m}) \times (45\,\text{Hz}) = 78.75\,\text{m s}^{-1}$$
- (ii) The tension T in the wire is:
$$T = \mu v^2 = (4.0 \times 10^{-2}\,\text{kg m}^{-1}) \times (78.75\,\text{m s}^{-1})^2 = 0.04 \times 6201.56 = 248.06\,\text{N} \approx 248\,\text{N}$$

Practice Question 7 (Harmonic Resonance in a Closed Pipe)

A pipe $20\,\text{cm}$ long is closed at one end. Which harmonic mode of the pipe is resonantly excited by a $430\,\text{Hz}$ source? Will the same source be in resonance with the pipe if both ends are open? (Take the speed of sound in air as $340\,\text{m s}^{-1}$).

(A) Fundamental (first harmonic) mode; Yes, it will also resonate when both ends are open.

(B) Fundamental (first harmonic) mode; No, it will not resonate when both ends are open.

(C) Third harmonic mode; No, it will not resonate when both ends are open.

(D) Second harmonic mode; Yes, it will also resonate when both ends are open.

Correct Answer: (B)

Worked Solution:

For the pipe of length $L = 20\,\text{cm} = 0.20\,\text{m}$ closed at one end, the fundamental frequency is:

$$\nu_1 = \frac{v}{4L} = \frac{340\,\text{m s}^{-1}}{4 \times 0.20\,\text{m}} = 425\,\text{Hz}$$

Higher odd harmonics are $3\nu_1 = 1275\,\text{Hz}$, $5\nu_1 = 2125\,\text{Hz}$, etc. Thus, the $430\,\text{Hz}$ source resonantly excites the **fundamental (first harmonic) mode**.

If both ends are open, the fundamental frequency becomes $\nu_1^{\text{open}} = \frac{v}{2L} = \frac{340}{2 \times 0.20} = 850\,\text{Hz}$, and higher harmonics are $1700\,\text{Hz}, 2550\,\text{Hz}, \ldots$. Therefore, the $430\,\text{Hz}$ source **will not be in resonance** with the open pipe.

BOSONANCE

Practice Question 8 (Assertion–Reason: Speed of Sound in Gases)

Assertion (A): At constant temperature, the speed of longitudinal sound waves in humid air is greater than that in dry air, whereas changing the atmospheric pressure at constant temperature does not alter the speed of sound.

Reason (R): According to Laplace's formula $v = \sqrt{\frac{\gamma P}{\rho}}$, at constant temperature P/ρ remains constant when pressure changes, while the presence of water vapour reduces the effective density ρ of air.

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(C) (A) is true, but (R) is false.

(D) (A) is false, but (R) is true.

Correct Answer: (A)

Worked Solution:

For an ideal gas, $\frac{P}{\rho} = \frac{RT}{M}$. At constant temperature T, varying the pressure P changes the density ρ proportionally, leaving $v = \sqrt{\gamma P/\rho}$ unchanged. When moisture ($\mathrm{H_2O}$, molar mass $18\,\mathrm{g\,mol^{-1}}$) replaces dry air ($M \approx 29\,\mathrm{g\,mol^{-1}}$), the average molar mass M and density ρ decrease, causing v to increase. Hence both (A) and (R) are true and (R) correctly explains (A).

Practice Question 9 (Superposition & Resultant Amplitude)

Two harmonic travelling waves on a stretched string have the same amplitude $a = 4.0\,\mathrm{cm}$, the same frequency, and the same wavelength, and both travel in the $+x$ direction. If the phase difference between them is $\phi = \frac{2\pi}{3}\,\mathrm{rad}$, what is the amplitude of the resultant wave?

(A) $8.0\,\mathrm{cm}$

(B) $4.0\sqrt{3}\,\mathrm{cm}$

(C) $4.0\,\mathrm{cm}$

(D) $2.0\,\mathrm{cm}$

Correct Answer: (C)

Worked Solution:

When two co-propagating waves of equal amplitude a differ in phase by ϕ, the resultant wave has amplitude:

$$A(\phi) = 2a\cos\left(\frac{\phi}{2}\right)$$

Substituting $a = 4.0\,\mathrm{cm}$ and $\phi = \frac{2\pi}{3}\,\mathrm{rad}$:

$$A = 2 \times (4.0\,\mathrm{cm}) \times \cos\left(\frac{\pi}{3}\right) = 8.0\,\mathrm{cm} \times \frac{1}{2} = 4.0\,\mathrm{cm}$$

BOSONANCE

Practice Question 10 (Statement-Based: Progressive vs. Stationary Waves)

Read the following two statements carefully:

Statement I: In a harmonic progressive wave of a given frequency, all particles of the medium oscillate with the same amplitude, but their phases differ from one position to another at any given instant.

Statement II: In a stationary wave, all particles located between two consecutive nodes oscillate in the same phase at a given instant, but have different amplitudes depending on their position.

Choose the correct option:

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct, but Statement II is incorrect.

(D) Statement I is incorrect, but Statement II is correct.

Correct Answer: (A)

Worked Solution:

In a progressive wave $y(x,t) = a\sin(kx - \omega t)$, every particle has the same amplitude a, while its phase $(kx - \omega t)$ varies continuously with position x. In a stationary wave $y(x,t) = (2a\sin kx)\cos\omega t$, all particles between two consecutive nodes share the same sign of $\sin kx$ and vibrate in unison ($\cos\omega t$) with zero phase difference between them, while their amplitude $|2a\sin kx|$ varies from 0 at the nodes to $2a$ at the antinode. Both statements are completely accurate.

7. Common NEET Pitfalls & Quick Revision Checklist

- **Pitfall 1 — Confusing Wave Velocity with Particle Velocity:** Never equate $v = \omega/k$ with $v_p = a\omega\cos(kx - \omega t)$. Wave speed v is constant in a uniform medium, whereas particle speed v_p oscillates harmonically between 0 and $a\omega$.
- **Pitfall 2 — Unit Mismatch in $y(x,t)$ Equations:** Often in exam problems, x and y are given in cm while t is in s. Always check units of k (cm^{-1} vs. m^{-1}) before computing wavelength $\lambda = 2\pi/k$ or wave speed $v = \omega/k$.
- **Pitfall 3 — Harmonic vs. Overtone Numbering in Closed Pipes:** For an open pipe or stretched string, the 1st overtone is the 2nd harmonic ($2\nu_1$). For a **closed organ pipe** (where even harmonics are missing), the 1st **overtone is the** 3rd **harmonic** ($3\nu_1$), and the 2nd overtone is the 5th harmonic ($5\nu_1$)!
- **Pitfall 4 — Phase Change on Reflection of Displacement vs. Pressure Wave:** At a rigid/closed boundary, the *displacement* wave undergoes a phase change of π (forming a displacement node), whereas the *pressure/density* wave reflects with zero phase change (forming a pressure antinode).
- **Pitfall 5 —Amplitude of Standing Wave vs. Component Wave:** In $y(x,t) = 2a\sin(kx)\cos(\omega t)$, the maximum amplitude at an antinode is $2a$, whereas the amplitude of each superposing travelling wave is half that value (a).

8. Concise Final Recap

1. Mechanical waves require an elastic, inertial medium and transfer energy without net mass flow. Transverse waves require shear rigidity (solids/stretched strings); longitudinal waves require bulk elasticity (solids, liquids, gases).
2. A progressive harmonic wave is described by $y(x,t) = a\sin(kx \mp \omega t + \phi)$, where $k = \frac{2\pi}{\lambda}$, $\omega = 2\pi\nu = \frac{2\pi}{T}$, and wave speed $v = \frac{\omega}{k} = \nu\lambda$.
3. Wave speeds: $v = \sqrt{\frac{T}{\mu}}$ (transverse wave on string); $v = \sqrt{\frac{B}{\rho}}$ (longitudinal wave in fluid); $v = \sqrt{\frac{Y}{\rho}}$ (solid rod); $v = \sqrt{\frac{\gamma P}{\rho}} = \sqrt{\frac{\gamma RT}{M}}$ (sound in ideal gas with Laplace's adiabatic correction).
4. Waves reflect with a phase reversal of π at a rigid boundary and zero phase change at an open boundary.
5. Superposition of opposite-travelling identical waves forms standing waves $y(x,t) = 2a\sin(kx)\cos(\omega t)$ with nodes and antinodes spaced $\lambda/2$ apart (and adjacent node-antinode distance $\lambda/4$).

6. Normal modes: Stretched string and open organ pipe support all harmonics $\nu_n = \frac{nv}{2L}$ ($n = 1, 2, 3, \ldots$); closed organ pipe supports only odd harmonics $\nu = \left(n + \frac{1}{2}\right) \frac{v}{2L}$ ($n = 0, 1, 2, \ldots$).
7. Two co-propagating sound waves of slightly different frequencies ν_1 and ν_2 interfere temporally to produce beats at frequency $\nu_{\text{beat}} = |\nu_1 - \nu_2|$.

BOSONANCE